



Department of Electrical & Computer Engineering
Rajshahi University of Engineering & Technology

Course Plan / Outline

1. **Course Information:**

Course Title	Numerical Methods & Discrete Mathematics Sessional
Course Code	ECE 2214
Course Type	Program Course (Sessional)
Credit	1.50 (Three Contact Hours per Week), three lecture slots per week according to class routine.
Semester	2 nd Year Even Semester

2. **Instructor Information:**

Name of the Instructor	Md. Faysal Ahamed
Designation	Lecturer
Office Room	ECE 419
Cell	01632656045
Email	faysal.ahamed@ece.ruet.ac.bd
Webpage link	https://www.ece.ruet.ac.bd/faysalahamed

3. **Prerequisite:** None

4. **Course Rationale:**

This sessional course is a practical extension of the theoretical concepts covered in the corresponding lecture course. Through this sessional course, students will gain hands-on experience with the tools and techniques used in numerical analysis and discrete mathematics, including software tools for computation, graph analysis, and optimization. The sessional classes will provide opportunities for students to work on individual and group projects that involve solving mathematical problems using numerical methods and discrete mathematics techniques. These projects will enable students to gain a deeper understanding of the theoretical concepts and to develop critical thinking and problem-solving skills. Overall, this course is designed to provide students with a practical foundation for applying the principles and techniques of numerical analysis and discrete mathematics to real-world problems. By the end of the course, students will have gained proficiency in using relevant software tools and will have developed the skills required to solve complex engineering problems in various fields. The sessional course complements the theoretical understanding gained in the lecture course and prepares students for further study in these areas.

5. **Course Objectives:**

After successfully completing this course, students will-

1. Develop proficiency in using relevant software tools for numerical computation, graph analysis, and optimization.
2. Gain hands-on experience with the tools and techniques used in numerical analysis and discrete mathematics, including interpolation, numerical differentiation and integration, and iterative methods for solving equations.
3. Solve mathematical problems using numerical methods and discrete mathematics techniques, both individually and in groups, and develop critical thinking and problem-solving skills.
4. Gain experience in interpreting and presenting mathematical results effectively, both in written and oral form.

5. Apply the principles and techniques of numerical analysis and discrete mathematics to real-world problems in various fields, including engineering, computer science, and the physical and life sciences.

Course Content / Course Description:

Sessional based on the theory of course ECE 2213. Course content of ECE 2213 is as follows:

Modeling, Computers and Error Analysis: Mathematical modeling and engineering problem solving, Programming and software, Approximations and round-off errors, Truncation errors and the Taylor series.

Roots of Equations: Bracketing Methods, Open Methods, Roots of polynomials. Linear Algebraic Equations: Gauss elimination, LU Decomposition and Matrix Inversion, Gauss-Seidel.

Curve Fitting: Least-square Regression. Interpolation, Interpolation with one and two independent variables, Formation of different difference table, Newton's forward and backward difference, LaGrange's interpolation.

Numerical Integration: Newton-Cotes Integration formulas, Integration of equations.

Optimization: One-dimensional Unconstrained optimization.

Numerical Differentiation and Differential Equations: Runge-Kutta Methods, Boundary-Value and Eigenvalue problems, Numerical solution of partial differential equations

Set: Operations on Sets, Algebraic Properties of Set, Computer Representation of Set, the Power Set Theorem.

Relation: Property of Relation, Binary Relations, Partial Ordering Relations, Equivalence Relations.

Propositional Logic: Syntax, Semantics, Valid, Satisfiable and Unsatisfiable Formulas, Encoding and Examining the Validity of Some Logical Arguments, Predicate and Quantifier, Universal and Existential Quantification; Modus Ponens and Modus Tollens.

Proof Techniques: The Structure of Formal Proofs, Direct Proofs, Proof by Contradiction, Mathematical Induction

Number Theory: Theorem of Arithmetic, Modular Arithmetic, GCD, LCM, Prime Number, Congruence, Application of Congruence, Application of Number Theory, Chinese Remainder Theory.

Introduction to Counting: Basic Counting Techniques - Inclusion and Exclusion, Pigeon Hole Principle, Permutation, Combination.

Mapping Course Learning Outcomes (COs) and Program Learning Outcomes (POs):

CO No.	CO Statement	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Apply numerical methods to solve analytical problems, including computational techniques for optimization, and equation solving.		C3										
CO2	Analyze proficiency in discrete mathematics concepts such as logic, and set theory use related software tools for analysis and problem-solving.					C4							

COs Assessment Policy:

COs	Lab report	Lab final
CO1	√	√
CO2	√	√

References:

SL No.	Text Book/Reference Book(s)	Author(s)	Online Availability
1.	Introductory Methods of Numerical Analysis	S.S. Sastry	Link is provided in Google Class Room
2.	Numerical Methods for Engineers	Steven C. Chapra & Raymond P. Canale	Link is provided in Google Class Room

3.	Computer Oriented Numerical Method	V. Rajaraman	Link is provided in Google Class Room
4.	Numerical Methods	Balagurusamy	Link is provided in Google Class Room
5.	Discrete Mathematics and Its Applications	Kenneth H. Rosen	Link is provided in Google Class Room

10.

Course plan specifying content, COs, co-curricular activities (if any), teaching learning assessment strategy mapped with COs:

Week	Topic	Delivery Material	Teaching Learning Strategy	Assessment Strategy	Aligned COs
1	Overview of the MATLAB software, review the basics, how to create .M file, edit window, command window, Basic input and output, looping etc.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO1
2	Write a MATLAB program to find out the roots of any non-linear equation by using Bisection Method, False Position Method, NR Method.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO1
3	Write a MATLAB program to find out the roots of any non-linear equation by using Secant Method. Write a MATLAB program to find out the roots of sets of any algebraic equations by using Gauss elimination method, Jacobi Method.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO1
4	Write a MATLAB program to find out the roots of sets of any algebraic equations by using Gauss-Jordan Method. Write a MATLAB program to find out the roots of sets of any algebraic equations by using Gauss-Seidal Method. Write a MATLAB program to find out the roots of sets of any algebraic equations by using Gauss-Seidal Acceleration Method.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO1

5	Write a MATLAB program to create Newton's forward and backward difference table. Write a MATLAB program to predict the desired data from a set of data by using Newton's forward and backward interpolation. Write a MATLAB program to predict the desired data from a set of data by using extrapolation.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project ☒ Lab test ☐ Others	CO1
6	Write a MATLAB program to find out the volume of a cylinder by using Simpson's 1/3 rule. Write MATLAB program to fit proper curve from sets of data/equations by various methods.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project ☒ Demonstration ☒ Lab test ☐ Others	CO1
7	Write a program (in MATLAB/ C/ C++/ Python) to solve various set & algebraic operations.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project ☒ Demonstration ☒ Lab test ☐ Others	CO2
8	Write a program to examine the properties of binary relations, partial ordering relations, and equivalence relations.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project ☒ Demonstration ☒ Lab test ☐ Others	CO2
9	Write a code to construct and analyze propositional formulas using syntax and semantics such as Predicate and Quantifier, Universal and Existential Quantification.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project ☒ Demonstration ☒ Lab test ☐ Others	CO2

10	Use of proof techniques such as direct proof, proof by contradiction, and mathematical induction to solve problems in number theory.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO2
11	Explore basic counting techniques, including inclusion and exclusion, the pigeonhole principle, permutation, and combination.	Example codes	☒ Lab Experiments (Hands On Practice) ☐ Simulation/Emulation ☒ Lab Demonstration ☐ Open Ended Lab ☐ Mini-Project ☐ Audio/Video ☐ Others	☐ Quiz ☐ Board Viva ☒ Lab Report ☐ Presentation ☐ Project Demonstration ☒ Lab test ☐ Others	CO2
12.	Lab Final				

11. Mapping of Graduate Attributes (PO), Knowledge Profile (KP), Complex Engineering Problems (CEP) and Complex Engineering Activity (CEA)

Course Code and Title	Pos												KP				CEP				CEA
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PO1-PO2	PO3	PO5	PO6-PO8	PO4	PO1-PO7	PO10		
ECE 2214 Numerical Methods & Discrete Mathematics Sessional	✓				✓																

11. Virtual Availability of Lecture Materials:
All resources related to this course are available in online platform.

Medium	Google classroom/Zoom/MS Teams etc.
Instructions	
Class Code	

12. Assessment & Marks Distribution:

SL No.	Name of Assessment	Marks	Date